

BILIARY SYSTEM, SPLEEN & PANCREAS — Imaging (Radiology)

Overview

- **CT = mainstay** for inflammatory, neoplastic, traumatic conditions. **US = best for gallstones.** **MRI = no ionizing radiation.**

BILIARY SYSTEM — modalities

- **US = primary + best all-purpose** imaging.
- **Radionuclide (HIDA):** key role in **biliary atresia + cystic duct obstruction.**
- **CT:** shows stones, biliary dilatation, wall thickening — but expensive + radiation.
- **MRCP:** non-invasive, **has largely replaced diagnostic PTC/ERCP.**

US of GB & biliary system

- Stones **>1–2 mm** seen. CBD/CHD almost always seen (except distal end). **Normal intrahepatic ducts NOT seen** (only proximal few mm).
- **Normal CBD ≤ 7 mm. Normal wall < 3 mm. Fasting needed.**

Radionuclide (Tc-IDA)

- Fast 4h → IV Tc-labelled IDA. Normally **GB + CBD + duodenum** seen within **1 hour**. **Cystic duct obstruction = CBD + duodenum seen but GB NOT seen.**

ERCP

- Endoscope + catheter; **mainly therapeutic** now. **Complications: pancreatitis, perforation, bleeding, sepsis.**

PTC

- Alternative to ERCP; **better with dilated ducts.** **Complications: bleeding, septicaemia, biliary peritonitis.**

Gallstones & GB pathology

- **Gallstones:** 10–20% radio-opaque (plain film); **60–80% seen on CT.** **US: strongly echogenic foci + acoustic shadowing;** US not reliable for bile duct stones.
- **GB polyp:** mucosal echogenic focus, **NO posterior shadowing** (vs stone); mostly few mm; **>10 mm → indication for cholecystectomy.**
- **Acute cholecystitis (US):** thick wall **>3 mm**, distended **>40 mm**, sludge/stones, wall oedema, positive sonographic Murphy's sign. Wall oedema differentiates acute from chronic.

Jaundice — role of imaging

- Determine: obstructive or not (US/CT/MRCP), **site, cause** — commonly **CBD stone, carcinoma head of pancreas, ampullary.**
- **Obstructive jaundice → biliary dilatation;** may be **no intrahepatic dilatation** in first 48h; obstruction without dilatation in liver transplants.
- **US more sensitive than CT for biliary dilatation.** **ERCP/PTC = sometimes diagnostic, mainly therapeutic.** ("Double channel sign" = dilated CBD + pancreatic duct.)

SPLEEN

- **US: iso-echoic to liver.** CT/MRI good evaluation.
- **Common masses:** cyst (**hydatid**), abscess, **tumour** — **lymphoma > metastases.**

- **Splenomegaly >13 cm** (texture may be normal): **lymphoma, portal hypertension, chronic infection, haemolytic anaemia, leukaemia.**
- **Splenic trauma: most commonly injured organ in blunt trauma;** contusion/laceration/haematoma. US useful; **CT = standard** (also bleeding + other organ injury, esp. liver + left kidney).

PANCREAS

- **CT = main imaging (not affected by air).** US limited (absorbed by air). MRI/MRCP, angiography, ERCP occasional. **Normal duct 2 mm.**
- Oblique orientation; variable shape/size (measurements not useful); **feathery outline (CT),** medium/high echogenicity (US); atrophy common with age.

Pancreatic masses

- Carcinoma, lymph node, abscess/pseudocyst, focal pancreatitis.
- **Adenocarcinoma = commonest; 2/3 in head. Head tumour → early diagnosis (pancreatic + biliary duct dilatation); tail → late diagnosis (large + advanced).**
- **CA pancreas (CT):** focal mass deforming outline, **irregular obliteration of peripancreatic fat;** low-density tumour, lymph nodes, duct dilatation, liver involvement, **vascular encasement/thrombosis.**
- **Endocrine tumours:** small, don't deform outline → hard to detect; **hypervascular → selective angiography, dynamic CT/MRI, intraoperative US** show small round mass.

Pancreatitis

- **Acute:** variable (necrosis/haemorrhage/suppurative). **Pancreas enlarged (often diffuse), low-density/echogenicity areas (oedema + necrosis).** Imaging role: **stage (CT severity index/CTSI),** exclude complications (abscess, pseudocyst), exclude underlying carcinoma.
- **Pseudocyst:** thin/thick-walled cystic structure within/adjacent to pancreas, fluid-filled; **many resolve in 4–6 weeks;** US/CT follow-up.
- **Chronic (CT):** main pancreatic duct dilatation + beading, intraductal calcifications, pancreatic calcification, atrophy, pseudocysts.

High-yield one-liners

- **US best for gallstones (echogenic + acoustic shadowing); CT mainstay for inflammation/tumour/trauma; MRCP replaced diagnostic ERCP/PTC.**
- **Normal CBD ≤7mm, wall <3mm. GB polyp >10mm → cholecystectomy.**
- **Acute cholecystitis: wall >3mm + distended + sludge + Murphy's; wall oedema = acute vs chronic.**
- **Cystic duct obstruction (HIDA) = GB not seen but CBD + duodenum seen.**
- **Obstructive jaundice → biliary dilatation (US more sensitive than CT); may lag 48h.**
- **Spleen = most injured in blunt trauma (CT standard); splenomegaly >13cm; lymphoma > mets.**
- **Pancreatic adenocarcinoma = 2/3 head, early dx (duct dilatation); tail = late. Vascular encasement = unresectable.**
- **Chronic pancreatitis = duct dilatation/beading + calcification + atrophy. Pseudocyst resolves 4–6 wks.**

BILIARY, SPLEEN & PANCREAS IMAGING — Revision Table

Bold = high-yield. CT mainstay; US best for gallstones; MRCP replaced diagnostic ERCP/PTC.

BILIARY SYSTEM	
Topic	Key points
Modalities	US = primary/best all-purpose ; HIDA (biliary atresia, cystic duct obstruction); MRCP replaced diagnostic ERCP/PTC
US normals	Stones >1–2mm seen; normal CBD ≤7mm, wall <3mm ; intrahepatic ducts not seen; fasting needed
HIDA	GB+CBD+duodenum in 1h. Cystic duct obstruction = GB NOT seen, CBD+duodenum seen
ERCP	Mainly therapeutic; complications: pancreatitis, perforation, bleeding, sepsis
PTC	Better with dilated ducts; complications: bleeding, septicaemia, biliary peritonitis
Gallstones	10–20% radio-opaque, 60–80% CT ; US = echogenic + acoustic shadowing ; US unreliable for duct stones
GB polyp	Echogenic, NO shadowing ; >10mm → cholecystectomy
Acute cholecystitis (US)	Wall >3mm, distended >40mm, sludge/stone, wall oedema, +Murphy's ; wall oedema = acute vs chronic
Obstructive jaundice	Biliary dilatation (may lag 48h); US more sensitive than CT ; commonest = CBD stone, CA head pancreas
SPLEEN	
Topic	Key points
Imaging	US iso-echoic to liver; CT/MRI
Masses	Cyst (hydatid), abscess, tumour = lymphoma > metastases
Splenomegaly	>13cm ; lymphoma, portal HTN, chronic infection, haemolytic anaemia, leukaemia
Trauma	Most injured organ in blunt trauma ; CT standard (+ bleeding, liver/left kidney)
PANCREAS	
Topic	Key points
Imaging	CT main (not affected by air) ; US limited; normal duct 2mm ; feathery outline (CT)
Masses	Carcinoma, LN, abscess/pseudocyst, focal pancreatitis
Adenocarcinoma	Commonest; 2/3 in head (early dx — duct dilatation) ; tail = late/advanced ; CT: low-density, fat obliteration, vascular encasement = unresectable
Endocrine tumour	Small, don't deform outline; hypervascular → angiography/dynamic CT/MRI
Acute pancreatitis	Diffusely enlarged + low density (oedema/necrosis); CT to stage (CTSI) , exclude abscess/pseudocyst/carcinoma
Pseudocyst	Fluid cystic; many resolve 4–6 weeks ; US/CT follow-up
Chronic pancreatitis	Duct dilatation + beading, calcifications, atrophy, pseudocysts